Literal certificate verification

Twist index calculus · bounded verification round · 2026-10-25

One fixed system, two checked constructions

The precommitted degree/closure workload selects p37, seed 0, phi arm, system 0. First executions, independent coefficient/root audits and deterministic A/A checks pass for both constructions. This is one public synthetic polynomial system; the three presentations of scalar 271 are not independent secret samples.

Method	Working degree	Product degree / power	Search multiplications	Sparse-check multiplications
raw16	12	12 / 2	33,609,631	15,211
closure8	7	12 / 2	916,980	15,983

All counts are recorded Fp37 modular operations. Search additions are 33,300,873 and 896,786; inversion calls are 1,562 and 177. Inversion exponentiation products are included in multiplication counts. Both certificates independently check q squared; their multiplier polynomials need not be equal. The raw degree-4 through degree-8 prefix exactly matches the old record.

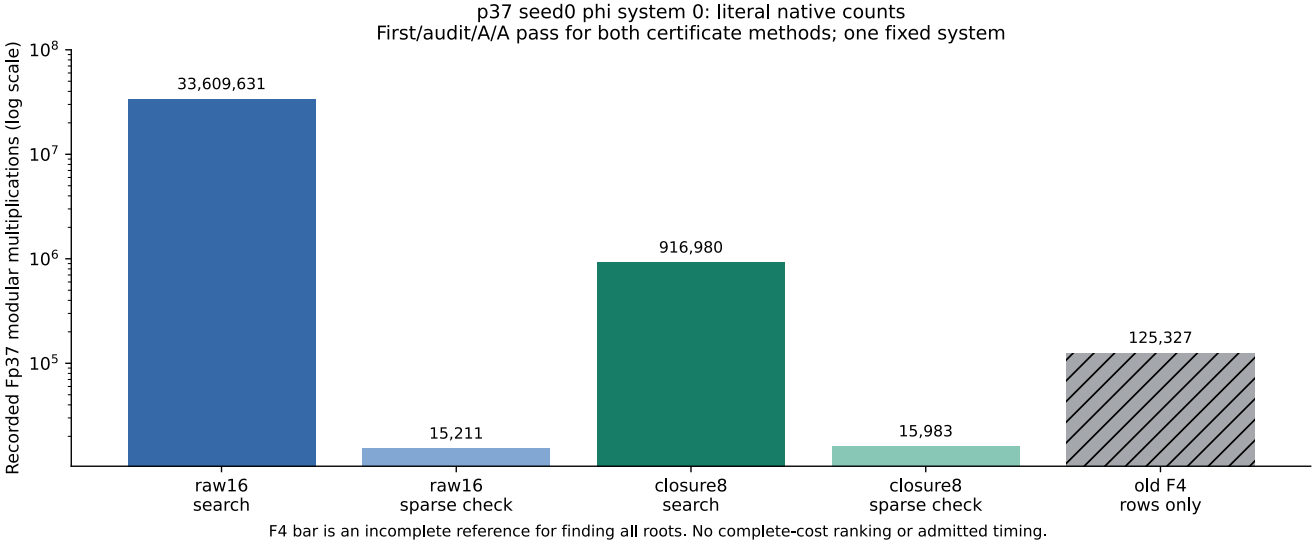


Figure 1: Recorded values from the two first records. Old F4 is a partial reference for the different task of finding all roots. No complete-cost ranking, admitted timing or fastest-tool claim follows.

Accounting boundary

The old F4 record has 125,327 row-reduction multiplications for this system. Its other work is unpriced. Even the closure search alone exceeds this partial count. Neither search ratio nor sparse verification cost establishes a complete filter or DLP improvement. Setup, hashes, allocation, memory, journaling, full F4 work, calibration and matched strongest IC/rho remain open. Callgrind is installed but no profile was executed in this round; client instructions alone would not close these obligations.

Exact identity and actual proof failures

Independent native integer lift

The exporter binds the original public input, equation packet, complete enumeration reference, passing prior audit, pair, peeled abscissa and raw8 prefix. It checks the retained finite root and the absence of an accepted physical relation in this source system. Fresh dense integer arrays check all 2,197 positions, independently of the constructor’s sparse verifier.

For $q = a^3 - a^2 s_1 + a s_2 - s_3$, every coefficient has zero remainder in

$$\sum h_i f_i = q^2 + 37R.$$

R has 454 nonzero coefficients. This integer lift, rather than a bare mod-37 assertion treated as a characteristic-zero identity, is the statement intended for the standard IDC checker. Statement SHA-256: `e7dd6d7622171f18c1d80ea9c1d26d182b08dbf664c3a3046115eb6715e5eb45`

Two attempts, both terminal

The native gates v1 and v2 each pass 13 tests: four instance controls and nine unchanged IDC tests. Format, build and source seals pass. Both actual exports pass and emit the same statement hash.

Attempt	Encoding	Lean exit	Maximum RSS
v1	Flat polynomial expressions	1: memory panic	3,775,120 KiB
v2	Balanced sums; four product lemmas	1: memory panic	3,772,400 KiB

Each actual Lean phase retained one aggregate 1,200-second limit and an 8 GiB address-space limit. RSS differs from address space. Durations are diagnostic on the shared host and are not admitted performance evidence. No third proof attempt was launched for this report. Neither attempt produced an accepted instance theorem, an axiom audit or an IDC. Failed proof sources and resource receipts remain preserved.

The theorem still owed

The literal integer identity must hold in an arbitrary commutative ring. Characteristic 37 must then give the q^2 identity; explicit absence of zero divisors and the unchanged generic cubic lemma must imply a = x, y or z at every common zero. No certificate identity may be assumed. All ten axiom sets must contain only propext, Classical.choice and Quot.sound. No sorry, native evaluation, custom axiom or kernel bypass is accepted. Lean is pinned to 4.34.1. lean4checker was not available.

The readable argument is conditional on the literal identity: common zeros annihilate every $h_i f_i$, hence $q^2 = 0$ in characteristic 37. Absence of zero divisors gives $q = 0$. The cubic identity gives $(a-x)(a-y)(a-z) = 0$, so one factor is zero. Formal compilation of the concrete identity remains incomplete; the native exact check is not a replacement for that proof.

Scope, evidence and remaining work

Relation finding and multiplicities

The current collector requires four distinct ordinary abscissae and one disjoint image abscissa. It rejects cubic roots that repeat internally or contain the peeled ordinary ID. The certificate targets rejection in that family. A repeated factor might encode a signed multiset or lower-arity relation after opposite signs cancel. Its folded row and contribution to rank have not been determined by this proof. An expanded audit must cover all 42 arms, preserve the old distinct set as a subset, combine multiplicities before deduplication and compare phi with doubling.

Original requirements retained

Requirement	Actual evidence / remaining gap
189 + 21 original cases	Complete; not repeated. No consistent twist-specific complete cold gain.
42 distinct zero-fiber arms	Complete first collection and independent audits. Factor-log rank remains the relevant output.
12 degree/closure configurations	Existing managed owner continues its full workload. Historical progress-v1 snapshot has 3 verified rows and 9 unverified/null rows.
42-arm certificate panel	Existing owner continues. Bounded failures are not nonexistence.
85 target configurations	Existing owner continues. No full recovery/cost conclusion is inferred here.
Concrete instance formal proof	Both bounded encodings failed from memory; original theorem remains open.
Complete end-to-end comparison	Unpriced work, matched strongest IC/rho, calibration and external comparison remain open.

The historical snapshot includes system-1 raw16: no certificate for powers 1,2,3 through degree 16 after 142,494,025 search multiplications. It exits 2 as partial. Later live records do not change this frozen snapshot. This is neither a completed panel nor a nonexistence proof.

Reproduction and provenance

Study: research/twist_index_calculus_20261020/zero_fiber_collector/ certificate_instance_proof. See REPORT.md, PROTOCOL.md, native/src/main.rs, run.sh, render.py and the retained metadata. Raw directories under the parent runs/: certificate-instance-native-gate-v1/v2 and certificate-instance-proof-v1/v2. The paired source records are under degree-closure-probe-panel-v1/cases/p37-seed0-phi-l0-active-sys0-raw16 and the corresponding closure8 directory. Original F4 reference: panel-v2/cases/p37-seed0/phi-l0-active/first/record.json.

Workload SHA-256: 254581dfebfcb992e8ce0c6b7896cbd7cb402259b0a5b72e34fd8c711cc0c88a Candidate SHA-256: 35d1890ef4aa04d3a7568dca950f85274097079a4813db64e8022025a57ba2ec Source freezes before exports: 379d5880d0 (v1), 7cc1b72419 (v2). The generic Lean prefix is copied unchanged. No source in any live collector was edited for this verification.

This report and its benchmark copy are unlisted. Canonical IC/rho graphs, leaderboard and benchmark index retain their previous data because complete price, reference and VALIDATION gates remain unmet. Literature/atlas matches place the method in known Semaev and degree-closure work; no novelty or atlas saturation is claimed. Tool documentation: official Callgrind manual.